Designing Intimate & Somatic Wearables for Menopause Fatigue

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Purpose

Perimenopause and menopause can bring 33+ symptoms into daily life, with 85% of women experiencing at least one. Among them, fatigue can feel diffuse and difficult to describe or quantify. **We are designing a soft, seamless wearable that senses the muscular signals associated with this exhaustion as it happens, transforming an invisible, hard-to-articulate experience into something people can see, understand, and use to better plan their cycles of work and rest.**

Gaps

- **Tracking tools** capture behavior, **not lived experience**: Dates, checklists, and yes/no questions miss the complexity & lived experience of fatigue.
- Menopause **research** remains **underfunded & stigmatized**: The 2002 Women's Health Initiative study contributed to decades of setbacks in menopause development.
- Fatigue is **studied in isolation.** Musculoskeletal, hormonal, & emotional factors are often studied separately. Difficult to understand or design for menopause-related exhaustion.

Background

47 million women enter menopause every year.

Symptoms arrive **across eight domains** at once: hot flashes, sleep, mood, memory, sexual health, urinary function, vaginal changes, and bodily pain.

Over **70% develop musculoskeletal symptoms**. **25% are disabled** by them. Aches and stiff joints rise 41% to 57%.

Underneath all of it runs exhaustion: diffuse, lingering for days, and heavy enough to reshape a working day. It is the symptom women find hardest to name.

- 24 weeks of estrogen deficiency costs 10% of strength and 18% of muscle fiber cross-section
- Muscle mass falls 0.6% per year after menopause
- Bone mineral density drops 10% during perimenopause
- Estrogen receptors concentrate on fast twitch (Type II) fibers, so power fades faster than strength

How Does Menopause Work?

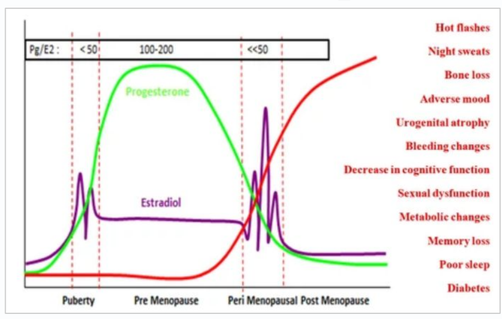


Fig. 1. *Hormonal fluctuations across the menopausal transition. [4]*

Perimenopause begins at a median age of 47.5 and lasts about four years. Menopause itself is a single day, confirmed only 12 months later.

Menopause unfolds in three stages:

Perimenopause | Menopause | Postmenopause

Women are born with a fixed supply of eggs + follicles (housing for eggs) & they decrease every year. As they decline, the feedback signal in body for hormones stager.

Progesterone goes first, Cycles begin skipping ovulation while periods look normal. Estrogen swings before it falls. Then the chain reaction. Falling estradiol sets off five changes at once: inflammation rises, bone density drops, joints degrade, muscle breaks down faster than it rebuilds, and the stem cells muscle relies on to repair itself become scarce.

Somatic Approach

Designing from the felt body rather than about it. Sensation & the journey is the material + data. **Bodies are never neutral,** and the differences between them are a resource to work with rather than noise to correct for.

How we are approaching it: **Sense first, ask second**. We are using our own journey and collecting user insight through workshops, journals and communication. From there we can design a wearable that is accurate and wanted. The wearable will read movement, touch and bodily rhythm continuously, then returns data as slow, intimate feedback rather than a graph to decode.

How it helps: User recognises her own signals early enough to act, and gains language for something that previously had none.
T**he outcome is not a score. It is a vocabulary.**



Fig. 2. *Provided kits with materials for our user study. Includes phone stand, markers, journal entries for multiple days and a bag*

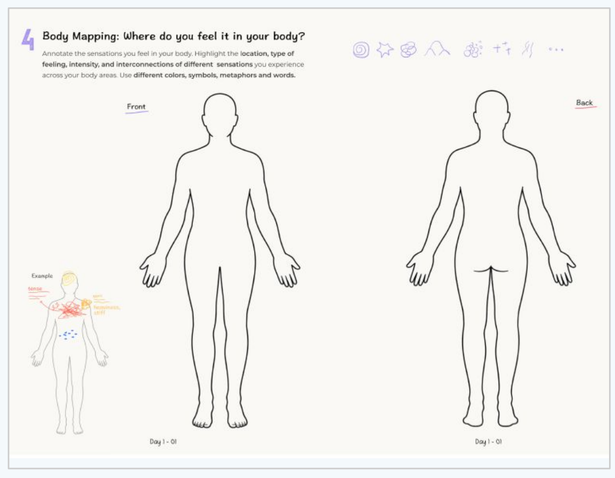


Fig. 3. *Body mapping page within our menopause fatigue journal.*

Methodology

Mapping Gaps

Literature across menopause endocrinology, musculoskeletal decline, cycle fatigue, soma design, and wearable sensing. Locate gaps.

Listen to Fatigue

Somatic workshops, body mapping and multi day journals. Participants mark where exhaustion sits, what it feels like, and when it arrives. The language for fatigue is built with participants.



Refine + Iterate

Probes return to the people who shaped them. Placement, material and feedback all iterate. Each pass sharpens two things at once: what the garment can sense, and what a wearer can name. Then the loop repeats.

Probe Prototypes

Sensor placement follows the body maps. Low fidelity probes are worn and tested against four things: common traits across participants, movement and posture, musculoskeletal signal, and felt fatigue as reported

User Feedback - Daily Journal

" I tuned into where underlying exhaustion manifests. Due to it not being prominent and the business of life, this exhibition was rejected for along time. Now I can understand it lays heavy under my eyes like a mask "

"The journal helped me feel more reflective about myself throughout the day and care more about my body from the additional time & thought being put in. I learned more about my body and the energy flow"

"Sometimes I felt more out of it before my period, now I understand that I feel more exhausted and there's a scientific reason behind it. I never felt to plan with my period in mind, but it would be very helpful to keep track of my hormones."

Body Mapping to Wearables:

Three directions, explored in parallel:

Expressive: The garment shows state on its own surface, keeping data attached to the body that produced it rather than migrating it to a screen. Prior work on soft exercise wearables found that summary graphs alienate data from experience and drive people to disengage over time.

Sensing: Continuous capture of movement and posture. The signal underneath the feeling.

Reactive. The garment acts back through pressure and touch.

Exploring and integrating different materials: Conductive fabric, conductive thread for soft sensing, IMUs for movement, inflatables for pneumatic pressure, textiles chosen for feel as much as function.

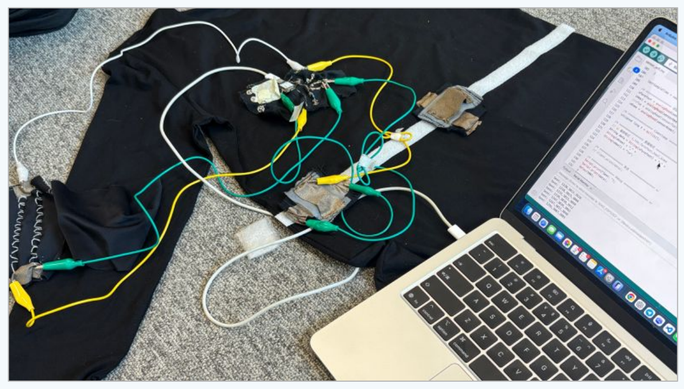


Fig. 4. *Prototyping different samples & types of wearables.*

Citations:

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